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CESIM – A Simulator for Clonal Evolution

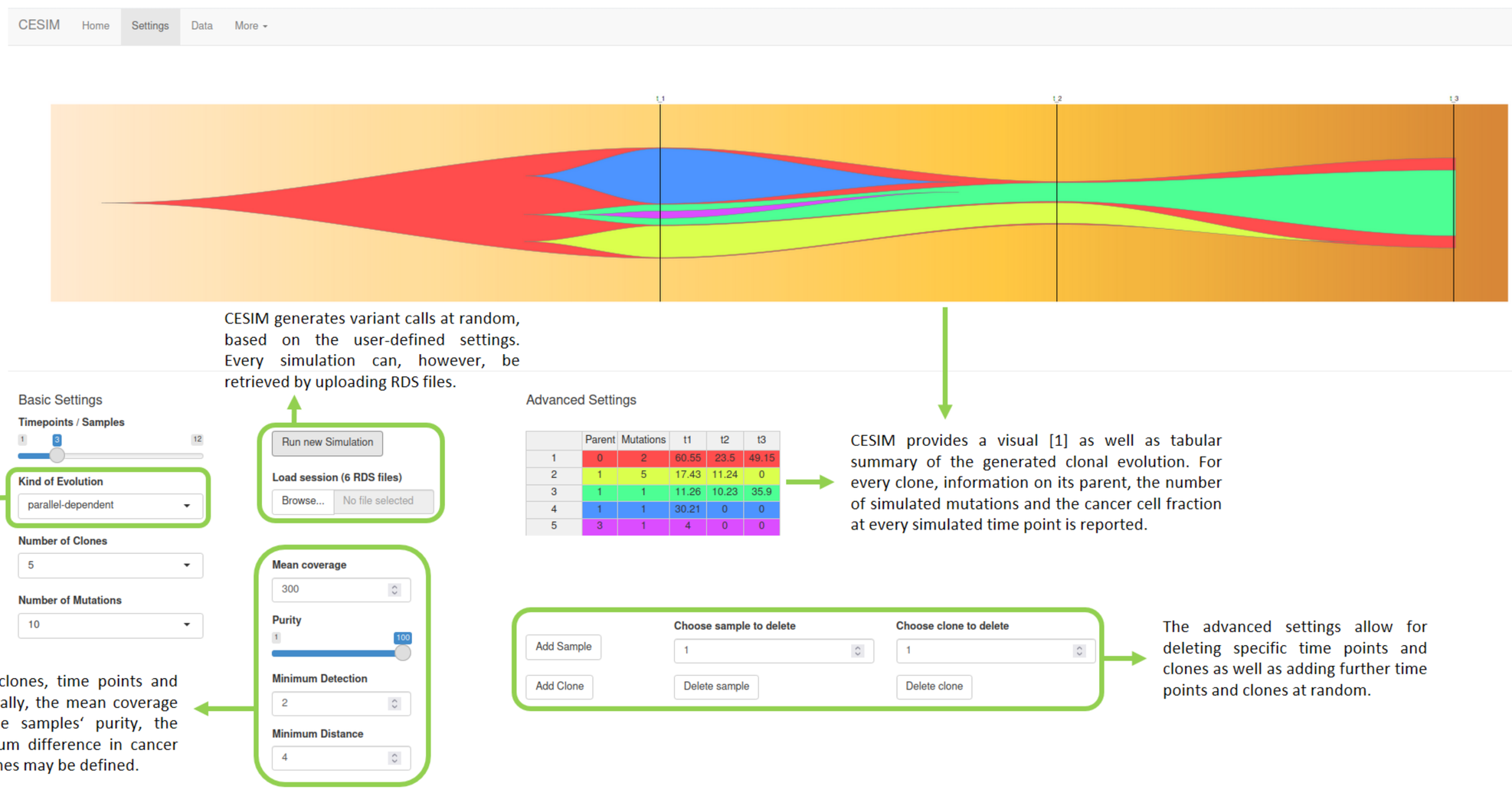
Background

Real data sets on clonal evolution, containing a variety of time points, different sources of data and evolution profiles, are sparse. Furthermore, ground truth is often unknown. Instead, the most likely clonal development is inferred from partly noisy, partly incomplete data. Therefore, realistic simulated data represent a key element in developing reliable algorithms for clonal evolution.

Methods and Results

We present CESIM – a Clonal Evolution SIMulator. Our novel simulation tool is able to generate realistic variant calls for an underlying user-definable clonal evolution. The output reported by CESIM can be taken as direct input for all common clonal evolution algorithms: 1) approaches performing clustering of variants, 2) reconstructing clonal evolution trees and 3) visualizing clonal evolution.

Settings:



CESIM generates variant calls at random, based on the user-defined settings. Every simulation can, however, be retrieved by uploading RDS files.

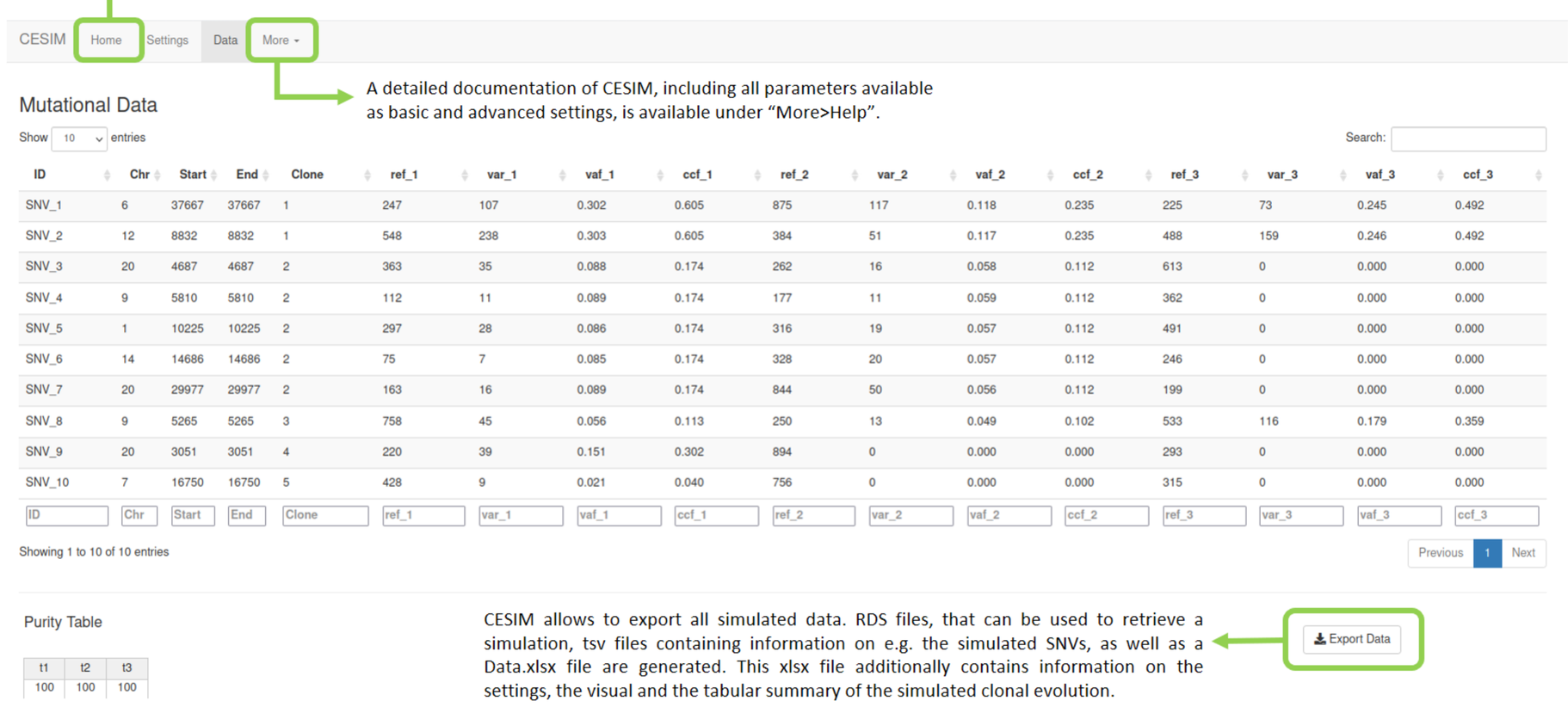
All main types of clonal evolution are considered: linear, branched dependent and branched independent (of note, punctuated and neutral evolution can be simulated as specific forms of linear and branched dependent evolution).

A user may select the number of clones, time points and mutations to be simulated. Additionally, the mean coverage of the sequencing experiment, the samples' purity, the detection threshold and the minimum difference in cancer cell fraction between two related clones may be defined.

CESIM provides a visual [1] as well as tabular summary of the generated clonal evolution. For every clone, information on its parent, the number of simulated mutations and the cancer cell fraction at every simulated time point is reported.

The advanced settings allow for deleting specific time points and clones as well as adding further time points and clones at random.

Data:



A short description on how to use CESIM is available under "Home".

A detailed documentation of CESIM, including all parameters available as basic and advanced settings, is available under "More>Help".

ID	Chr	Start	End	Clone	ref_1	var_1	vaf_1	ccf_1	ref_2	var_2	vaf_2	ccf_2	ref_3	var_3	vaf_3	ccf_3
SNV_1	6	37667	37667	1	247	107	0.302	0.605	875	117	0.118	0.235	225	73	0.245	0.492
SNV_2	12	8832	8832	1	548	238	0.303	0.605	384	51	0.117	0.235	488	159	0.246	0.492
SNV_3	20	4687	4687	2	363	35	0.088	0.174	262	16	0.058	0.112	613	0	0.000	0.000
SNV_4	9	5810	5810	2	112	11	0.089	0.174	177	11	0.059	0.112	362	0	0.000	0.000
SNV_5	1	10225	10225	2	297	28	0.086	0.174	316	19	0.057	0.112	491	0	0.000	0.000
SNV_6	14	14686	14686	2	75	7	0.085	0.174	328	20	0.057	0.112	246	0	0.000	0.000
SNV_7	20	29977	29977	2	163	16	0.089	0.174	844	50	0.056	0.112	199	0	0.000	0.000
SNV_8	9	5265	5265	3	758	45	0.056	0.113	250	13	0.049	0.102	533	116	0.179	0.359
SNV_9	20	3051	3051	4	220	39	0.151	0.302	894	0	0.000	0.000	293	0	0.000	0.000
SNV_10	7	16750	16750	5	428	9	0.021	0.040	756	0	0.000	0.000	315	0	0.000	0.000

CESIM reports variants together with information on the number of simulated reads with vs without the variant, the varianat allele frequency (vaf) and the cancer cell fraction (ccf) for every time point.

CESIM allows to export all simulated data. RDS files, that can be used to retrieve a simulation, tsv files containing information on e.g. the simulated SNVs, as well as a Data.xlsx file are generated. This xlsx file additionally contains information on the settings, the visual and the tabular summary of the simulated clonal evolution.

Discussion

CESIM is, to our knowledge, the first simulator for clonal evolution, generating realistic variant calls for a user-definable tumor development. Thereby, CESIM provides the basis for systematic evaluation of available approaches analyzing clonal evolution as well as development of new optimized pipelines.

References

[1] Miller, C.A. *et al.* (2016). Visualizing tumor evolution with the fishplot package for R. *BMC Genomics*, **17**(880).

Availability: https://imigitlab.uni-muenster.de/sandmans/cesim

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